# Analytic State-Averaged DMRG-CASSCF Gradients and Nonadiabatic Couplings in a PySCF/pyblock2 Workflow

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**Abstract**

Analytic nuclear gradients and nonadiabatic coupling vectors are central ingredients for locating conical intersections and for propagating nonadiabatic dynamics with multireference electronic-structure methods. Complete-active-space self-consistent-field (CASSCF) theory provides a robust reference near electronic degeneracies, but full configuration interaction in the active space limits the accessible number of active orbitals. Here we report a PySCF/pyblock2 workflow that connects DMRG active-space roots to PySCF's state-averaged CASSCF derivative infrastructure and to a SHARC-compatible output interface. A PySCF-compatible active-space solver wrapper supplies DMRG energies, density matrices, transition density matrices, root tracking, and response operations through the interfaces expected by PySCF. The workflow is validated against PySCF full-CI active-space references on public benchmark systems. We further quantify finite-bond-dimension response errors by replacing FCI active-space vectors with block2 matrix product states at fixed FCI-optimized orbitals. An ethylene interface regression test demonstrates that the DMRG-CASSCF backend writes the Hamiltonian, dipole, gradient, and derivative-coupling data blocks requested by SHARC.

# 1 Introduction

Nonadiabatic molecular processes require electronic-structure information beyond adiabatic energies. Surface-hopping algorithms and related mixed quantum–classical methods use gradients and state couplings to propagate nuclear motion through regions where the Born–Oppenheimer separation breaks down.[1–3] For same-spin internal conversion, analytic derivative coupling vectors are especially useful because they define, together with state gradients, the first-order topology near conical intersections and provide direct inputs to optimization and dynamics workflows.[4]

CASSCF and state-averaged CASSCF (SA-CASSCF) wave functions are widely used for this purpose because they provide a balanced multiconfigurational description of near-degenerate states. The conventional active-space solver, however, is a full configuration interaction (FCI) expansion whose dimension grows combinatorially with active electrons and orbitals. The density matrix renormalization group (DMRG) replaces the FCI vector by a matrix product state (MPS), making compact approximations to much larger active spaces possible when the entanglement structure is favorable.[5–7]

Analytic response theory for DMRG-CASSCF has developed rapidly. Freitag and coworkers introduced an approximate SA-DMRG-SCF gradient and nonadiabatic coupling formalism based on single-site MPS Lagrange multipliers in a mixed-canonical representation.[8] Iino, Shiozaki, and Yanai later formulated analytic nuclear gradients for SA-DMRG-CASSCF using newly derived coupled-perturbed equations.[9] These developments establish the theoretical foundation for DMRG-based derivative calculations. The remaining practical need addressed here is an open, reproducible PySCF-based implementation with regression tests, public benchmark data, and an interface usable by SHARC-style nonadiabatic dynamics workflows.

This work makes three contributions. First, we implement a PySCF-compatible DMRG active-space solver wrapper and accompanying response primitives that allow block2 DMRG roots to enter PySCF's existing SA-CASSCF gradient and nonadiabatic-coupling machinery.

Table 1: Positioning of the present work relative to closely related methods and software.

| Work or software | Scope | Relation to the present implementation |
|---|---|---|
| SA-CASSCF derivative couplings[4] | Analytic CASSCF NAC theory and implementations | Provides the FCI active-space response reference used for numerical validation. |
| SA-DMRG-SCF response[8] | Approximate analytic gradients and NACs with MPS response variables | Provides the theoretical DMRG response framework followed here. |
| SA-DMRG-CASSCF gradients[9] | Coupled-perturbed equations for analytic SA-DMRG-CASSCF gradients | Establishes an independent modern formulation; the present work emphasizes open PySCF/pyblock2 integration and NAC/interface validation. |
| PySCF and block2[10–12] | Open electronic-structure and DMRG infrastructure | Supplies the host CASSCF derivative code and the spin-adapted MPS backend. |
| SHARC[2,3] | Nonadiabatic dynamics driver and data interface | Defines the Hamiltonian, dipole, gradient, and derivative-coupling output contract tested in the ethylene regression case. |

Second, we validate gradients and nonadiabatic couplings against FCI active-space references and characterize the finite-bond-dimension response error on public benchmark systems. Third, we expose the validated backend through a SHARC-compatible PySCF interface and demonstrate the complete output path on a public ethylene test case. The contribution is therefore an implementation, validation, and reproducibility paper grounded in established SA-DMRG response theory and PySCF's CASSCF derivative infrastructure.

# 2 Theory and Algorithm

## 2.1 State-Averaged CASSCF Response

For an SA-CASSCF wave function, the orbitals and active-space CI coefficients are optimized for a weighted average of several electronic states. Analytic gradients and nonadiabatic couplings are obtained by differentiating a Lagrangian that enforces stationarity with respect to orbital rotations and active-space wave-function parameters. The resulting coupled-perturbed CASSCF equations determine the orbital-response multipliers that enter the effective one- and two-particle density matrices used in the derivative contractions.[4,8]

## 2.2 DMRG Active-Space Representation

In DMRG-CASSCF, each active-space state is represented by an MPS rather than by an explicit FCI vector. A finite bond dimension $M$ controls the rank of the MPS approximation and therefore the accuracy of energies, density matrices, gradients, and derivative couplings. Following the SA-DMRG-SCF response strategy of Freitag et al., the present implementation evaluates the response quantities needed by PySCF's SA-CASSCF derivative machinery while using pyblock2/block2 for the MPS representation and DMRG optimization.[8,12]

## 2.3 Bond-Dimension Response Error

To isolate the effect of the MPS compression, we define a fixed-orbital bond-dimension response error. For each benchmark system we first converge a standard SA-CASSCF calculation with an FCI active-space solver. We then hold the converged molecular orbitals fixed, run spin-adapted block2 DMRG in the same active-space Hamiltonian at a selected bond dimension $M$, convert the resulting MPS roots into the PySCF active-space representation, and recompute analytic gradients and nonadiabatic couplings. The error is measured against the FCI active-space derivative at the same orbitals,

$$\Delta_g(M) = \left\| \nabla_{\mathbf{R}} E_0(M) - \nabla_{\mathbf{R}} E_0^{\mathrm{FCI}} \right\|_2 , \tag{1}$$

$$\Delta_d(M) = \min_{\sigma=\pm 1} \left\| \mathbf{d}_{01}(M) - \sigma \mathbf{d}_{01}^{\mathrm{FCI}} \right\|_2 , \tag{2}$$

where the sign minimization removes the arbitrary global phase of the electronic states. This protocol avoids mixing the MPS compression error with nonstationarity from orbital reoptimization on a deliberately truncated CI manifold.

# 3 Implementation

The implementation is built on PySCF[10,11] and pyblock2/block2.[12] The DMRG-facing layer is designed to satisfy the solver and response contracts used by PySCF's `grad.sacasscf`, `nac.sacasscf`, `mcscf.newton_casscf`, and `grad.lagrange` modules.

Two code paths are provided. The `MPSAsFCISolver` path exposes a PySCF-compatible active-space solver interface, allowing DMRG roots to be used by PySCF's CASSCF gradient and nonadiabatic-coupling infrastructure. This path is the main regression-test target because it permits direct comparison with standard FCI active-space calculations. A native `CPDMRGCASSCFResponseMPS` response class is also included as an implementation path for MPS-routed response operations.

The workflow used in the public benchmarks is:

1. converge an SA(2)-CASSCF reference with PySCF and an FCI active-space solver;
2. freeze the converged orbitals and construct the active-space Hamiltonian in the same orbital basis;
3. optimize the target spin-adapted block2 MPS roots, optionally with a small buffer of additional candidate roots, at the selected bond dimension;
4. select and phase-align the target MPS roots by maximum overlap with the FCI reference;
5. evaluate energies, one- and two-particle density matrices, gradients, and derivative couplings through the PySCF response interface; and
6. compare the resulting derivatives to the FCI active-space values at identical orbitals.

For multistate calculations, the solver applies overlap-based root and phase continuity before returning CI vectors to PySCF. It may solve more candidate roots than are propagated and then returns only the target roots selected by overlap. In production calculations,

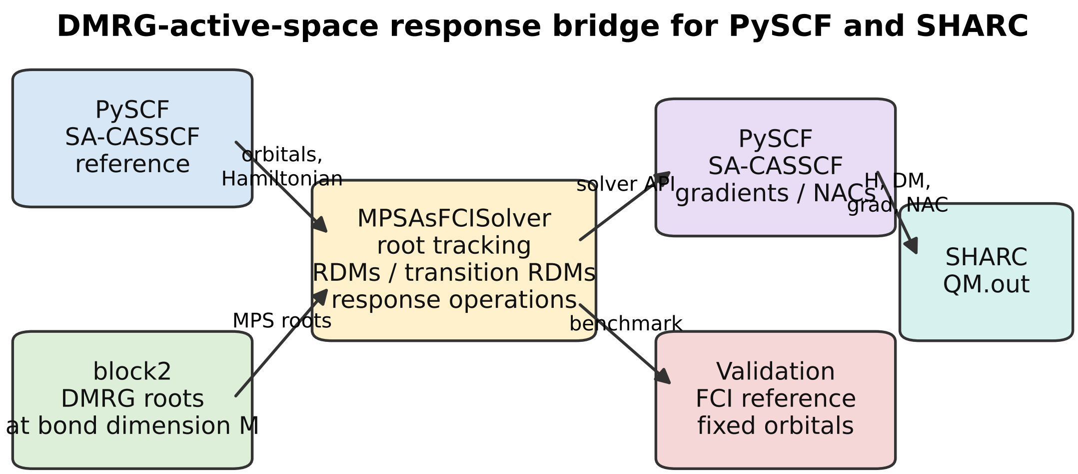


Figure 1: Software workflow used in the public validation and SHARC-interface tests. PySCF provides the SA-CASSCF derivative infrastructure, block2 provides DMRG active-space roots, and the DMRG-facing solver layer supplies the root tracking, density matrices, transition density matrices, and response operations required by the PySCF and SHARC-facing interfaces.

current DMRG roots are matched to the available reference from the previous CASSCF macro iteration or trajectory step by maximizing the absolute wave-function overlap, and each matched root is phase-aligned to that reference. No FCI reference is required for this production root following. The FCI references used below enter only the public validation protocol, where an absolute error relative to the exact active-space result is being measured.

The SHARC-facing interface adds a PySCF template method, `dmrg-casscf`. When requested, the interface constructs a PySCF CASSCF object, installs the DMRG-backed active-space solver, evaluates the requested Hamiltonian, dipole, gradient, and derivative-coupling quantities, and writes them in SHARC `QM.out` format. The ethylene example in this work is used as an end-to-end interface regression test.

# 4 Computational Details

All benchmark calculations used singlet state averaging over two roots with equal weights. The public systems are listed in Table 2. The bond-dimension response-error calculations used FCI-converged SA-CASSCF orbitals as the fixed orbital basis. Spin-adapted SU2 block2 DMRG was then run at each selected bond dimension, and the resulting target state vectors were selected and phase-aligned to the FCI reference before evaluating derivative errors. The same root-buffer assignment machinery used by the production interface was enabled for dense or symmetry-related candidate subspaces. Gradients are reported in m$E_\mathrm{h}$/Bohr and derivative couplings in atomic units.

The FCI reference calculations used restricted Hartree–Fock orbitals converged to $10^{-12}$, SA-CASSCF energy convergence of $10^{-10}$, and a CASSCF gradient convergence threshold of $10^{-7}$. The bond-dimension scan used up to 30 DMRG sweeps and a sweep tolerance of $10^{-12}$ unless noted otherwise. The ethylene SHARC-interface regression test used an SA(2)-DMRG-CASSCF(2,2)/STO-3G model with $M = 60$, ten DMRG sweeps, a DMRG sweep tolerance of $10^{-8}$, CASSCF energy convergence of $10^{-8}$, and CASSCF gradient convergence of $10^{-5}$. These settings are intentionally small enough for a public regression test and are not used to make a mechanistic chemistry claim.

# 5 Results and Discussion

## 5.1 Validation Against FCI Active-Space References

The implementation is covered by unit and integration tests for DMRG/FCI energies, one- and transition-density matrices, MPS-to-FCI conversion, analytic gradients, analytic derivative couplings, and SHARC output formatting. The current regression suite contains 46 passing tests across 14 test files. These tests verify both direct numerical agreement with FCI active-space references in small active spaces and consistency between the native MPS

Table 2: Public validation and benchmark systems.

| System | Basis sets | Active space | Role |
|---|---|---|---|
| $H_4$ chain | STO-3G, 3-21G, 6-31G | CAS(4,4) | Small-system response validation |
| $H_2O$ | STO-3G, 3-21G, 6-31G | CAS(4,4)/CAS(6,6) | Small polar molecule benchmark |
| $N_2$ | STO-3G, 3-21G, 6-31G | CAS(6,6) | Medium-active-space response validation |
| $C_2$ | STO-3G, 3-21G, 6-31G | CAS(8,8) | Larger active-space/root-tracking benchmark |
| LiF | STO-3G, 3-21G, 6-31G | CAS(4,4) | Nonzero-NAC avoided-crossing benchmark |
| Ethylene | STO-3G, 3-21G, 6-31G | CAS(2,2) | SHARC-interface and $\pi/\pi^*$ benchmark |
| Butadiene | STO-3G, 3-21G, 6-31G | CAS(4,4) | Conjugated $\pi$-system benchmark |
| Formaldehyde | STO-3G, 3-21G, 6-31G | CAS(4,4) | Carbonyl excited-state benchmark |
| Benzene | STO-3G, 3-21G, 6-31G | CAS(6,6) | Aromatic $\pi$-space benchmark |

and PySCF-compatible solver paths.

## 5.2 Finite-Bond-Dimension Response Error

Figure 2 shows representative fixed-orbital bond-dimension response-error curves. The selected systems cover small active spaces, CAS(6,6) and CAS(8,8) examples, nonzero derivative couplings, and a public SHARC-interface molecule. The curves show systematic reduction of the derivative error as $M$ is increased and recover the FCI active-space limit at high $M$. The high-$M$ gradient errors in Table 4 are reported in m$E_\mathrm{h}$/Bohr; values near $10^{-4}$ in this column correspond to $10^{-7}$ $E_\mathrm{h}$/Bohr and are well below the 0.1 m$E_\mathrm{h}$/Bohr reference line in Figure 2. The NAC column is an absolute vector error and is interpreted together with the benchmark matrix in Figure 3.

The fixed-orbital protocol also separates active-space truncation from orbital relaxation. With real DMRG roots evaluated at the same FCI-stationary orbitals, the response errors

Table 3: Validation layers included in the public test and benchmark suite.

| Validation target | Public evidence | Quantity checked |
|---|---|---|
| DMRG active-space solver | Solver regression tests | Energies, state density matrices, transition density matrices, and analytic NAC pipeline consistency with FCI references. |
| MPS/FCI mapping | Active-space map tests | Round-trip mapping for CAS(2,2), CAS(4,4), and CAS(8,8) active spaces. |
| Response linear algebra | Coupled-perturbed response tests | Hermiticity, linearity, response-vector solves, and agreement between MPS and FCI response operations in exactly recoverable cases. |
| Gradient and NAC derivatives | Derivative regression tests | Analytic gradients and derivative couplings against PySCF FCI active-space references. |
| SHARC interface | Ethylene interface regression | Presence of `H`, `DM`, `GRAD`, and `NACDR` blocks in `QM.out` with SHARC master error code 0. |

follow smooth bond-dimension convergence instead of the large excursions produced by an earlier SVD-truncated CI-vector proxy.

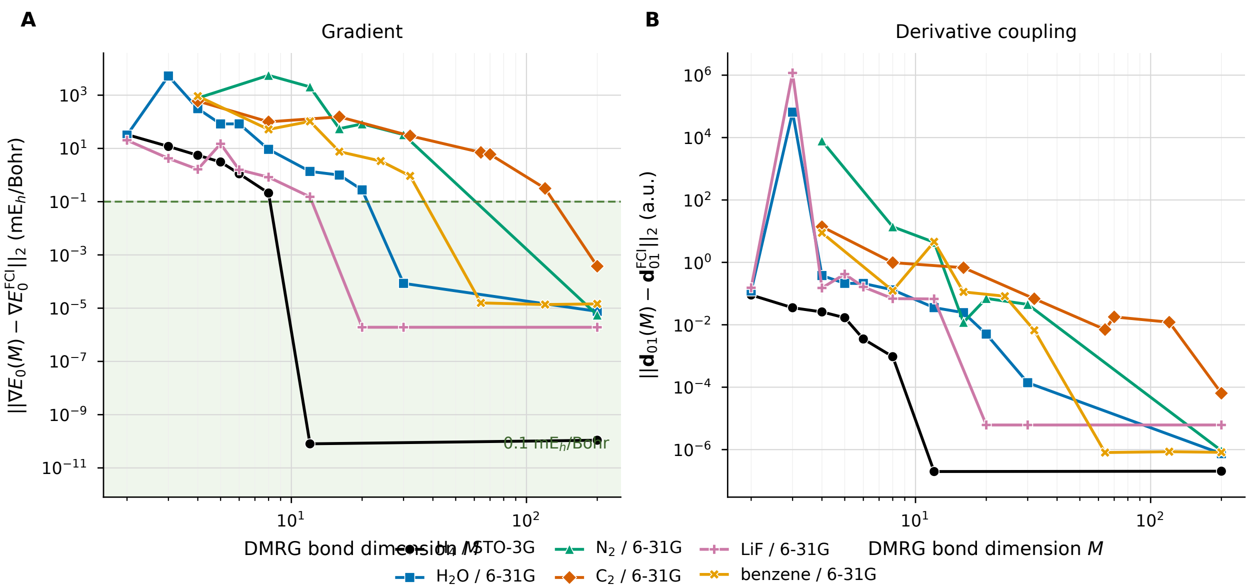


Figure 2: Fixed-orbital bond-dimension response-error convergence for public benchmark systems. Gradient errors are reported in m$E_\mathrm{h}$/Bohr and nonadiabatic coupling errors in atomic units.

### 5.3 SHARC Interface Regression Test

The public ethylene test requests the Hamiltonian, dipole matrices, gradients for both singlet states, and the $S_0/S_1$ derivative coupling. The calculation completes with SHARC master error code 0 and writes all requested data blocks to `QM.out`. This confirms that the

Table 4: Representative largest-$M$ fixed-orbital response errors from the public benchmark scan. The full molecule/basis matrix is summarized in Figure 3.

| System | Active space | $M$ | $\Delta_g$ (m$E_\mathrm{h}$/Bohr) | $\Delta_d$ | Role |
|---|---|---|---|---|---|
| $H_4$/STO-3G | CAS(4,4) | 200 | $1.10 \times 10^{-10}$ | $2.03 \times 10^{-7}$ | High-$M$ exact-rank check |
| $H_2O$/6-31G | CAS(6,6) | 200 | $7.60 \times 10^{-6}$ | $7.42 \times 10^{-7}$ | Polar molecule benchmark |
| $N_2$/6-31G | CAS(6,6) | 200 | $5.52 \times 10^{-6}$ | $9.24 \times 10^{-7}$ | Medium CAS benchmark |
| $C_2$/6-31G | CAS(8,8) | 200 | $3.76 \times 10^{-4}$ | $6.40 \times 10^{-5}$ | Larger active-space benchmark |
| LiF/6-31G | CAS(4,4) | 200 | $1.92 \times 10^{-6}$ | $6.16 \times 10^{-6}$ | Avoided-crossing benchmark |
| Ethylene/6-31G | CAS(2,2) | 200 | $5.63 \times 10^{-6}$ | $7.46 \times 10^{-7}$ | SHARC-interface benchmark |
| Butadiene/6-31G | CAS(4,4) | 200 | $2.28 \times 10^{-4}$ | $1.02 \times 10^{-4}$ | Conjugated $\pi$ benchmark |
| Formaldehyde/6-31G | CAS(4,4) | 200 | $3.19 \times 10^{-6}$ | $1.16 \times 10^{-6}$ | Carbonyl benchmark |
| Benzene/6-31G | CAS(6,6) | 200 | $1.45 \times 10^{-5}$ | $8.17 \times 10^{-7}$ | Aromatic $\pi$ benchmark |

DMRG-backed PySCF object satisfies the data contract expected by the SHARC interface for energies, dipoles, gradients, and derivative couplings.

# 6 Conclusions

We have implemented analytic SA-DMRG-CASSCF gradients and nonadiabatic couplings in a PySCF/pyblock2 workflow and exposed the backend through a SHARC-compatible PySCF interface. The implementation is validated against FCI active-space references, and public benchmark data quantify how gradient and derivative-coupling errors decay with DMRG bond dimension at fixed FCI-optimized orbitals. The results provide a reproducible open-source platform for testing DMRG-based multireference response quantities and for connecting them to nonadiabatic dynamics interfaces.

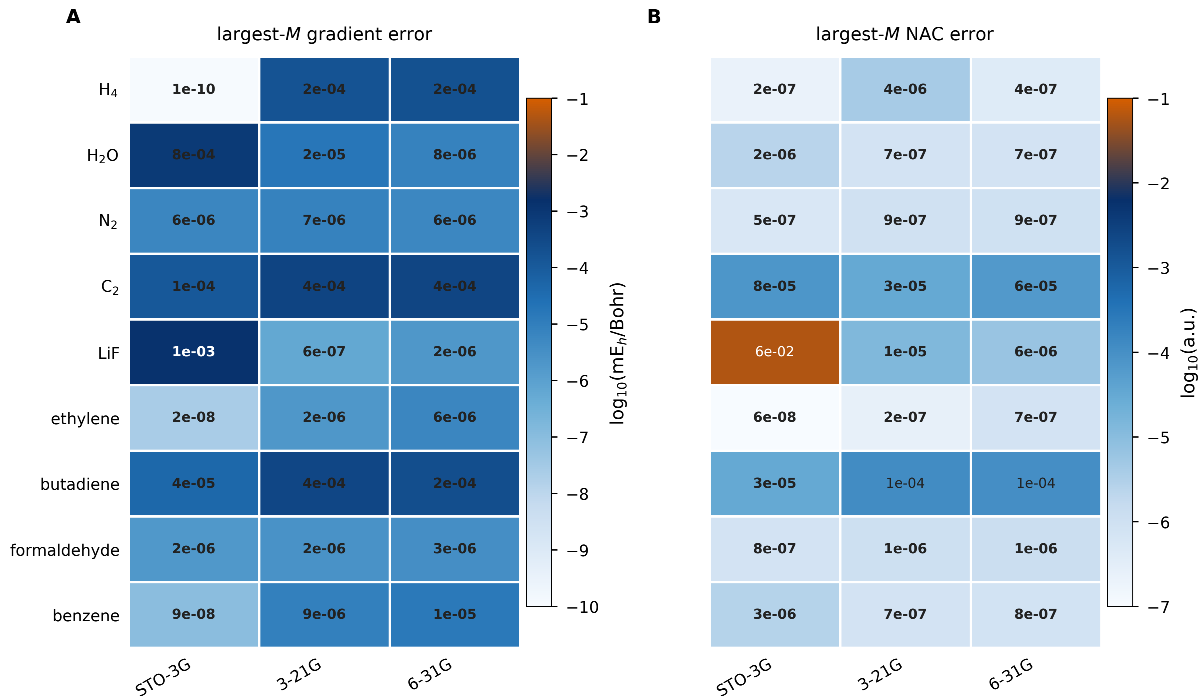


Figure 3: Largest-$M$ derivative errors across the public molecule/basis matrix. Color encodes the base-10 logarithm of the error, while each cell reports the corresponding gradient error in m$E_\mathrm{h}$/Bohr or NAC error in atomic units.

# Supporting Information

Supporting Information: benchmark inputs, full convergence tables, regression-test summary, and SHARC ethylene interface example files.

# Data and Code Availability

The public code bundle contains the response implementation, tests, benchmark drivers, benchmark summaries, plotting scripts, and the ethylene SHARC interface regression example. The repository is available at `https://github.com/yurika1030sakura/dmrg_sacasscf`. A permanent archival DOI will be added before final publication. Application systems and trajectory data outside the public benchmark set are not included.

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